

Week 9

Wednesday 3/18

Boundary Value Problems

these problems are of the form:

$$y'' = f(x, y(x), y'(x))$$

$$y(a) = \alpha, \quad y(b) = \beta$$

on $[a, b]$ (spatial)

methods:

1) The Linear Shooting Method

2) Finite Diff method

3) Finite Element method

1) The Linear Shooting Method:

i) main idea

ii) systems of 1st order ODEs

iii) Applying 1st order method to 1st order system

iv) the shooting method

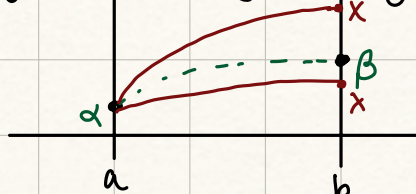
i) main idea: shooting method. We want to solve the linear BVP

$$y'' - p(x)y' - q(x)y = r(x)$$

$$y(a) = \alpha, \quad y(b) = \beta$$

→ lean on theory for IVP

$$y(a) = \alpha, \quad y'(a) = y'_a$$



"shoot" to hit β - wrong slope $\rightarrow$ miss

* solve the problem w/ the "correct" slope *

if solution to IVP is found such that $y(b) = \beta$ then how do we begin to solve using previous methods

ii) Systems of 1st order ODEs

2nd-order ODEs (linear or nonlinear) can always be written as a system of 1st order ODEs

$$\rightarrow y'' = f(x, y, y'), \quad y(a) = \alpha_1, \quad y'(a) = \alpha_2$$

$$\rightarrow \text{substitution: } y_1 = y; \quad y_1' = y' = y_2$$

$$\vec{y} = \vec{f}(x, \vec{y}) \begin{cases} y_1' = y_2 \\ y_2' = f(x, y_1, y_2) \end{cases}$$

$$y_1(a) = \alpha_1, \quad y_2(a) = \alpha_2$$

ex transform $y'' - p(x)y' - q(x)y = r(x)$ w/ $y(a) = \alpha$ $y'(a) = \alpha_2$

$$\rightarrow y_1' = y_2 \quad y_2' = p(x)y_2 + q(x)y_1 + r(x) \quad \} \quad \vec{y}' = f(x, \vec{y})$$

iii) Apply 1st order methods for $y' = f(x, y)$, $y(a) = \alpha$
to solving $\vec{y}' = \vec{f}(x, \vec{y})$, $\vec{y} = \vec{\alpha}$

ex apply Eulers to

$$y_1' = y_2^2 - 2y_1$$

$$y_2' = y_1 - y_2 - t y_2^2$$

Euler's method: $\vec{w}_{i+1} = \vec{w}_i + h f(t_i, \vec{w}_i)$

$$\begin{aligned} \rightarrow y_1(t_i) &\approx w_{1,i} \\ \rightarrow y_2(t_i) &\approx w_{2,i} \end{aligned} \rightarrow \begin{bmatrix} w_{1,i+1} \\ w_{2,i+1} \end{bmatrix} = \begin{bmatrix} w_{1,i} \\ w_{2,i} \end{bmatrix} + h \begin{bmatrix} (w_{2,i}^2 - 2w_{1,i}) \\ (w_{1,i} - w_{2,i} - t_i w_{2,i}^2) \end{bmatrix}$$

iv) the linear shooting method
nonhomogeneous eqn:

$$y_p'' - p(x)y_p' - q(x)y_p = r(x)$$

homogeneous eqn:

$$y_h'' - p(x)y_h' - q(x)y_h = 0$$

general solution:

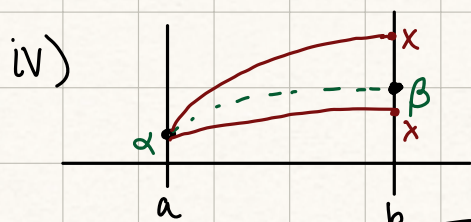
$$y(x) = y_p(x) + \int y_h(x)$$

$\rightarrow$ this is a solution for any s

Friday 3/20

the linear shooting method

- i) main idea
- ii) systems of 1st order ODEs
- iii) Applying 1st order method to 1st order system
- iv) the shooting method



$y(x) = y_p(x) + s y_h(x)$ HW: show this soln for any $s \in \mathbb{R}$ need to satisfy:

$$y(a) = y_p(a) + s y_h(a) = \alpha$$

$$y(b) = y_p(b) + s y_h(b) = \beta$$

nonhomogeneous eqn: $y_p(a) = \alpha$

$$y_p'' - p(x) y_p' - q(x) y_p = r(x)$$

homogeneous eqn: $y_h(a) = 0$

$$y_h'' - p(x) y_h' - q(x) y_h = 0$$

$$y'(a) = \underbrace{y_p'(a)}_0 + s \underbrace{y_h'(a)}_1$$

$$s = \frac{\beta - y_p(b)}{y_h(b)}$$

note about existence & uniqueness

- no soln. if $y_h(b) = 0$ and $y_p(b) \neq \beta$
- if $y_h(b) = 0$ and $y_p(b) = \beta$, then there $\exists$ infinite number of solutions

Outline of Shooting Method

→ solve $y'' + p(x)y' - q(x)y = r(x)$
 $y(a) = \alpha, \quad y(b) = \beta$

i) By replacing this w/ two IVP

$$y'' - p(x)y' - q(x)y = r(x) \quad y(a) = \alpha, \quad y'(a) = 0$$

$$y'' - p(x)y' - q(x)y = 0 \quad y(a) = 0, \quad y'(a) = 1$$

ii) Convert each into systems

$$y_{p1} = y_p \quad \tilde{y}_{p1} = y'_p = y_{p2}$$

$$y'_{p1} = y_{p2}$$

$$y'_{p2} = p(x)y'_{p2} + q(x)y_{p1} + r(x)$$

$$y_{p1}(a) = \alpha \quad y_{p2}(a) = 0$$

repeat w/ $y_{h1} = y_h \quad y'_{h1} = y_{h2}$

$$\rightarrow y(x) = y_{p1}(x) + \frac{\beta - y_{p1}(b)}{y_{h1}(b)} y_{h1}(x)$$

round off error:

If $y_p \gg \beta$, then $y(x) \approx y_p - \frac{y_{p1}(b)}{y_{h1}(b)} y_{h1}(x)$

if $x \leq b$, then

$$y(x) \approx y_{p1}(x) - y_{p1}(x)$$

possible cancellation & loss of digits